ESTR1002 Fall 2025

Lab Exercise #5

Due: 15 Oct 2025

Separate your answer for each question in a different .c file. Compress all files into a .zip and submit it to Blackboard.

Question 1

Write a C program to compute the value of "n chooses k", defined as $\binom{n}{k} = \frac{n!}{k!(n-k)!}$ where $m! = m \times (m-1)!$ and $0! = 1$. The program asks for two positive integers n and k from the user, prints out the value of "n chooses k", and then ends the program. You may assume the value of n and k is always less than or equal to 12.

Input: two integers n and k, where 0 <= k <= n <= 12.

Output: the value of "n chooses k".

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
6 2
15
```

Sample Run #2:

```
5 5
1
```

Question 2

Ask for an integer N from the user and print out an N by N matrix on screen. The first element of the matrix is 1. For all other elements, it has to be greater than the element on left (if any) and the element above (if any) by 1.

Input: an integer N in [1, 10].

Output: the matrix described above, where consecutive elements in the matrix are separated by a comma and then a **tab**.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
4
1,    2,    3,    4
2,    3,    4,    5
3,    4,    5,    6
4,    5,    6,    7
```

Sample Run #2:

```
6
1,    2,    3,    4,    5,    6
2,    3,    4,    5,    6,    7
3,    4,    5,    6,    7,    8
4,    5,    6,    7,    8,    9
5,    6,    7,    8,    9,    10
6,    7,    8,    9,    10,   11
```

Question 3

For an array consisting of n elements, index ii in [0,n-1] is said to be an equilibrium index, if the sum of elements in sub-array A[0...i-1] equals to the sum of elements in sub-array A[i+1...n-1] (excluding the element at index i).

At the two ends, 0 is said to be an equilibrium index if A[1] + A[2] + … + A[n-1] = 0 and n-1 is said to be an equilibrium index if A[0] + A[1] + … + A[n-2] = 0.

Given an array of integers, say A, you need to output all the equilibrium indices of A. If A does not have any equilibrium index, output -1.

Input: there are two lines of inputs.

- The first line is an integer N representing the length of array A, where 3 <= N <= 20000.
- The second line is an array A containing N integers, separated by whitespace.

Output: print out all the equilibrium indices of array A, in ascending order. All indices are separated by whitespace. If no equilibrium index is found , print out -1.

Beware that a native implementation may take a long time to run. Make sure that your program can be finished within a second for large cases.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
3
-2 5 -2
1
```

Sample Run #2:

```
7
-7 1 5 2 -4 3 0
3 6
```

Sample Run #3:

```
3
1 3 5
-1
```

Question 4

Given an array p with n number of integers. The array alternates between counting up and counting down. A turning point is defined as an element where the direction of the sequence changes from increasing to decreasing or from decreasing to increasing. For example, in the array [1, 2, 5, 6, 2, 1, 3, 5, 7, 5, 3, 2, 4, 6], the turning points are '6', '1', '7', and '2'.

Input: the inputs have two lines. The first line contains the number n ($1 \leq n \leq 2000$), representing the length of the array p. The second line contains n space-separated integers representing the elements of the array p.

Output: the sum of all turning points.

The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
6
1 2 3 2 1 2
4
```

The turning points are 3 and 1.

Sample Run #2:

```
10
3 2 5 8 6 9 8 7 6 9
31
```

The turning points are 2, 8, 6, 9, and 6.

Sample Run #2:

```
5
1 2 2 2 1
0
```

There is no turning point.

Question 5

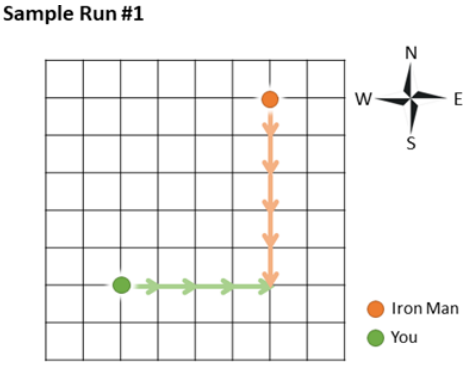
Today, your favorite film star --- Iron Man, will be touring our city, and you want to get a picture with him. Iron Man's tour will go on a pre-defined path along the streets of the city, and you know the exact path. Your goal is to meet him at one of the crossroads on his path as early as possible.

The streets running north-south and east-west meet together, forming an infinite grid. Iron Man's tour starts at the crossroad **X** blocks east and **Y** blocks north of the crossroad where you are initially located. For each minute, both you and he can only walk through exactly one block in any of the 4 directions (north, east, south, or west) to the next crossroad. While Iron Man moves to the next crossroad as planned, you can choose to stand still for a minute, or to walk to the next crossroad.

You need to meet Iron Man at the same crossroad at the same time before the end of his tour. Is it possible to get a picture with him? If so, how soon can you do it?

Input: each test case (Sample Run) has two lines. The first line has two integers **X** and **Y in** [-1000, 1000]. The second line contains an integer **L** followed by a list of **L** integers, where **L** <= 2000 is the total number of minutes of Iron Man's tour, and the list of **L** integers records the move direction in every minute. Each integer in the list could be 1, 2, 3, or 4, which denotes direction 'E', 'W', 'S', or 'N', respectively.

Output: an integer, representing the least number of minutes needed to get a picture with Iron Man. If there is no way to get a picture, print "impossible". To help you better understand the problem, the figure shown below illustrates the input and solution of the "Sample Run #1", where you get the picture by first moving east four times, and then waiting there for one minute.



The sample output is shown as follows (the underlined characters are the inputs from the user).

Sample Run #1:

```
4 5
5 3 3 3 3 3
5
```

Sample Run #2:

```
0 -1
6 3 4 1 3 3 1
2
```

Sample Run #3:

```
2 10
6 4 3 4 4 3 4
impossible
```